

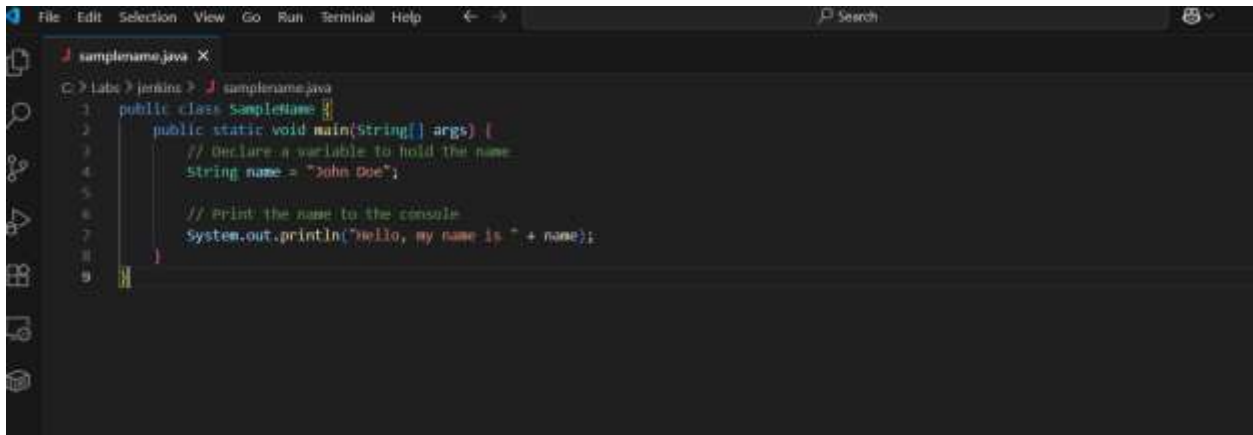
Experiment-5

Demonstrate continuous integration and development using Jenkins

AIM :

The objective of this experiment is to utilize Jenkins, a widely used automation server, to configure and execute a pipeline script. This script will run a sample Java program, demonstrating Jenkins' capability to automate build and execution processes in a CI/CD environment.

1-Create Samplejava file



```
File Edit Selection View Go Run Terminal Help
samplename.java X
C:\labs\jenkins> samplename.java
1 public class SampleName {
2     public static void main(String[] args) {
3         // Declare a variable to hold the name
4         String name = "John Doe";
5
6         // Print the name to the console
7         System.out.println("Hello, my name is " + name);
8     }
9 }
```

```
Admin@DESKTOP-A5IG8HL MINGW64 /c/labs/jenkins (Main)
$

Admin@DESKTOP-A5IG8HL MINGW64 /c/labs/jenkins (Main)
$ git init
Initialized empty Git repository in C:/Labs/jenkins/.git/

Admin@DESKTOP-A5IG8HL MINGW64 /c/labs/jenkins (master)
$ git add .

Admin@DESKTOP-A5IG8HL MINGW64 /c/labs/jenkins (master)
$ git commit -m "samplename.java"
[master (root-commit) 05d3877] samplename.java
1 file changed, 9 insertions(+)
create mode 100644 samplename.java
```

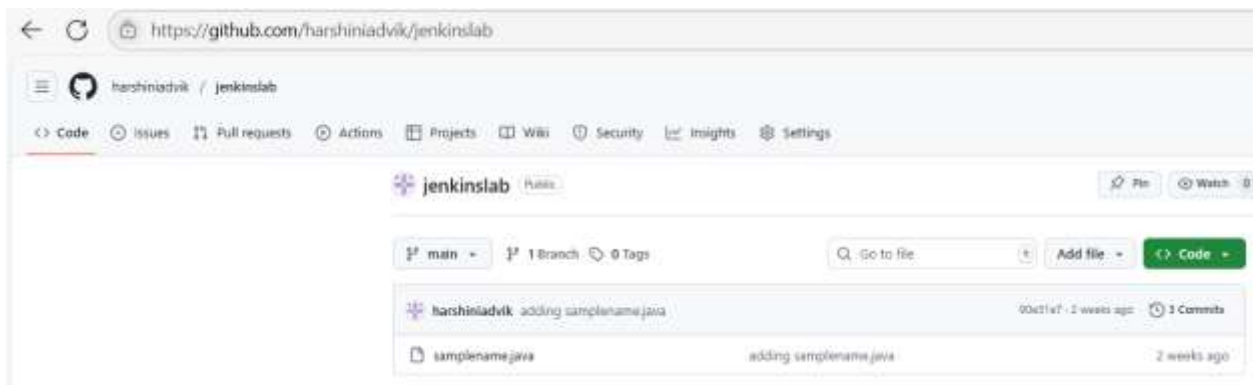
```
Admin@DESKTOP-A5IG8HL MINGW64 /c/labs/jenkins (master)
$ git config user.email "v.harshini1817@gmail.com"

Admin@DESKTOP-A5IG8HL MINGW64 /c/labs/jenkins (master)
$ git config user.name "harshiniadvik"

Admin@DESKTOP-A5IG8HL MINGW64 /c/labs/jenkins (master)
$ git status
On branch master
nothing to commit, working tree clean
```

Use Command : **Git push**

and check the samplejava file in git browser



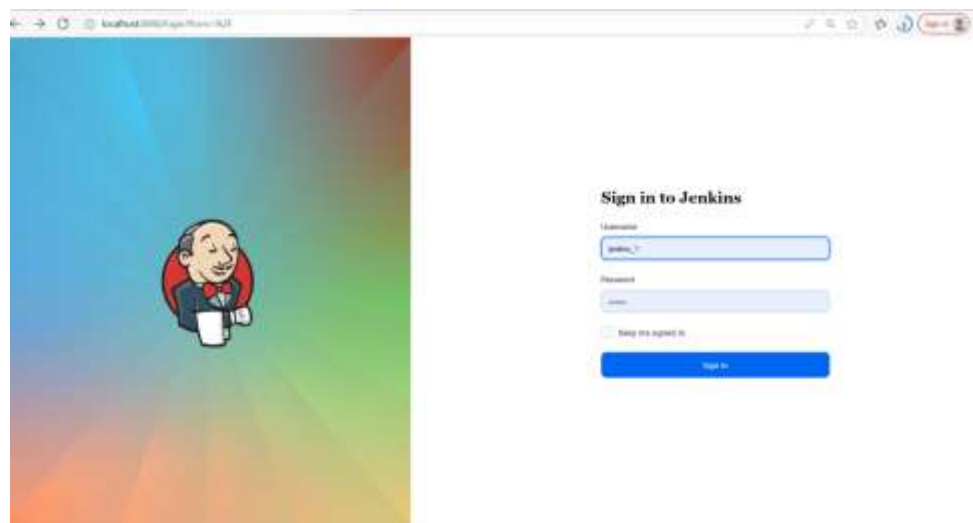
As we have worked on jenkins in previous experiment-4 now try to run the war file from downloads again and login to jenkins

->Go to jenkins war path and run **java -jar jenkins.war**

```
C:\Users\Admin\Downloads>java --version
java 17.0.12 2024-07-16 LTS
Java(TM) SE Runtime Environment (build 17.0.12+8-LTS-286)
Java HotSpot(TM) 64-Bit Server VM (build 17.0.12+8-LTS-286, mixed mode, sharing)

C:\Users\Admin\Downloads>java -jar jenkins.war
Running from: C:\Users\Admin\Downloads\jenkins.war
webroot: C:\Users\Admin\Downloads\jenkins.war
2025-07-22 09:35:44.926+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2025-07-22 09:35:45.041+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2025-07-22 09:35:45.162+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.0.19; built: 2025-04-0
1T13:28:40.005Z; git: 21436cd577b4238238eb192683dfc2b1476c6b87; jvm 17.0.12+8-LTS-286
2025-07-22 09:35:46.179+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /
, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2025-07-22 09:35:46.251+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2025-07-22 09:35:46.878+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\Admin\
Downloads\jenkins found at: $user.home/.jenkins
2025-07-22 09:35:47.848+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$Co
reContextHandler@1cfd1875{Jenkins v2.504.2,/,b=file:///C:/Users/Admin/.jenkins/war/,a=AVAILABLE,h=oeje9n.ContextHandler$
CoreContextHandler$CoreToNestedHandler@28c0b664[STARTED]}
2025-07-22 09:35:47.871+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@1deb2c43
[HTTP/1.1, (http/1.1)]{0.0.0.0:8080}
2025-07-22 09:35:47.889+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@280aa59a[ST
ARTING][12.0.19,sto=0] @2923ms
2025-07-22 09:35:47.889+0000 [id=27] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPor
t=disabled
2025-07-22 09:35:47.385+0000 [id=26] INFO jenkins.model.Jenkins#<init>: Starting version 2.504.2
2025-07-22 09:35:47.482+0000 [id=33] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2025-07-22 09:35:48.149+0000 [id=32] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
```

Login with credentials if old credential works we can use it else create new one as per the previous experiment and login



Go to dashboard

and click on newitem

Then provide a new name for the project and select **pipeline** in the **item type** as per the below image

New Item

Enter an item name
jenkinsci

Select an item type:

- Freestyle project**
Classic, general purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Pipeline**
Orchestrates long running activities that span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.
- Multi-configuration project**
Template for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
- Multibranch Pipeline**
Creates a set of Pipeline projects according to detected branches in one SCM repository.
- Organisation Folder**
Creates a set of multibranch project subfolders by scanning for repositories.

On next screen we need to configure in the pipeline

Dashboard | jenkinsci | Configuration

Configure

General Enabled

Description

How can I [discards](#)?

☒ Discard old builds : 0

☐ Do not allow concurrent builds

☐ Do not allow the pipeline to resume if the controller restarts

☐ Stop on success

☐ Pipeline speed/throughput override : 0

☐ Preserve status from completed builds : 0

☐ This project is parameterized : 0

☐ Rebuild builds : 0

Triggers

Set up automated actions that start your build based on specific events, file code changes or scheduled times

☒ Build after other projects are built : 0

☐ Build periodically : 0

☐ Branch source trigger for (ST)cm polling : 0

☐ Poll SCM : 0

☐ Trigger builds remotely (e.g. from scripts) : 0

Pipeline

Define your Pipeline using Groovy directly or pull it from version control

Definition

Pipeline script

Script : 0

```
1 pipeline {
2   agent any
3   stages {
4     stage('Build') {
5       steps {
6         checkout scm
7       }
8     }
9   }
10 }
```

[View](#) [Apply](#)

localhost:8080/job/jenkinslab/configure

Configure

General

Build

Triggers

Build

Advanced

Triggers

Let us recommend actions that start your build based on specific events like code changes or scheduled times

☐ Build when other projects are built

☐ Build periodically

☐ GitHub hook trigger for GIT polling

☐ Poll SCM

☐ Trigger builds remotely (e.g. via scripts)

Pipeline

Define your Pipeline using Groovy directly or pick a Pipeline template

Definition

Pipeline script

```

1 script {
2   agent any // Use any available agent (cloud/master) for this build
3
4   checkout scm
5
6   gitBranches {
7     gitBranches {
8       // Recommended with your SCM integration
9       getBranches('*/*/**/*', '*/**/*')
10    }
11  }
12
13  stages {
14    stage('Hello World') {
15      steps {
16        // Recommended with your SCM integration
17        getBranches('*/*/**/*', '*/**/*')
18      }
19    }
20  }
21 }

```

☒ Use Groovy Syntax

Pipeline Syntax

Advanced

Advanced

Save Apply

In the pipeline script copy and paste the below script in the box and save it:

```

pipeline {
    agent any // Use any available agent (node/worker) for the build

    environment {
        GIT_URL = 'https://github.com/harshiniadvik/jenkinslab.git' // Replace with your Git URL
        GIT_BRANCH = 'main' // Replace with the branch you want to checkout
    }

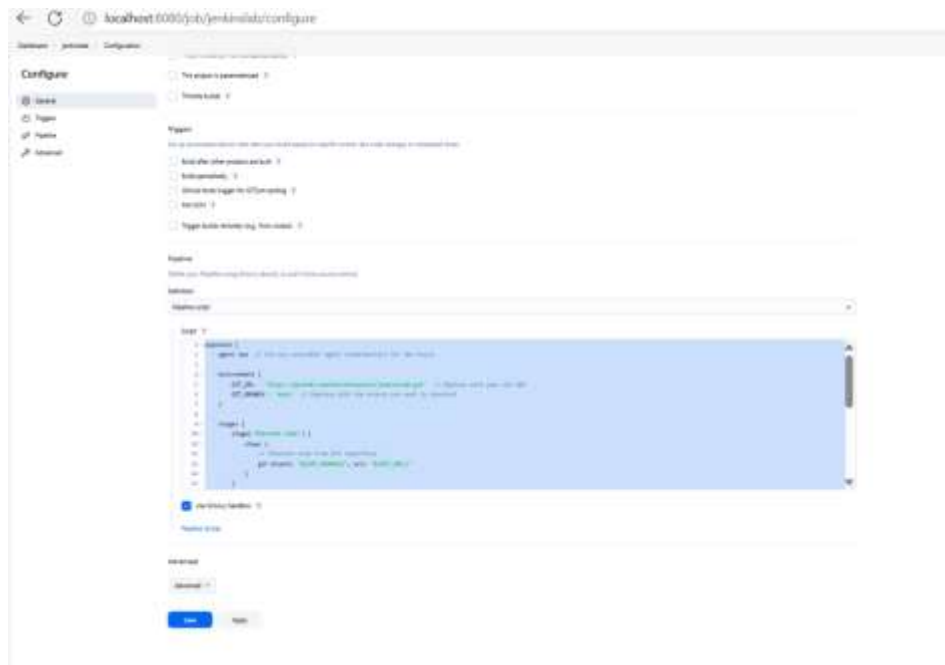
    stages {
        stage('Checkout Code') {
            steps {
                // Checkout code from Git repository
                git branch: "${GIT_BRANCH}", url: "${GIT_URL}"
            }
        }

        stage('Build') {
            steps {
                bat 'javac Samplename.java'
                bat 'java Samplename'
            }
        }
    }

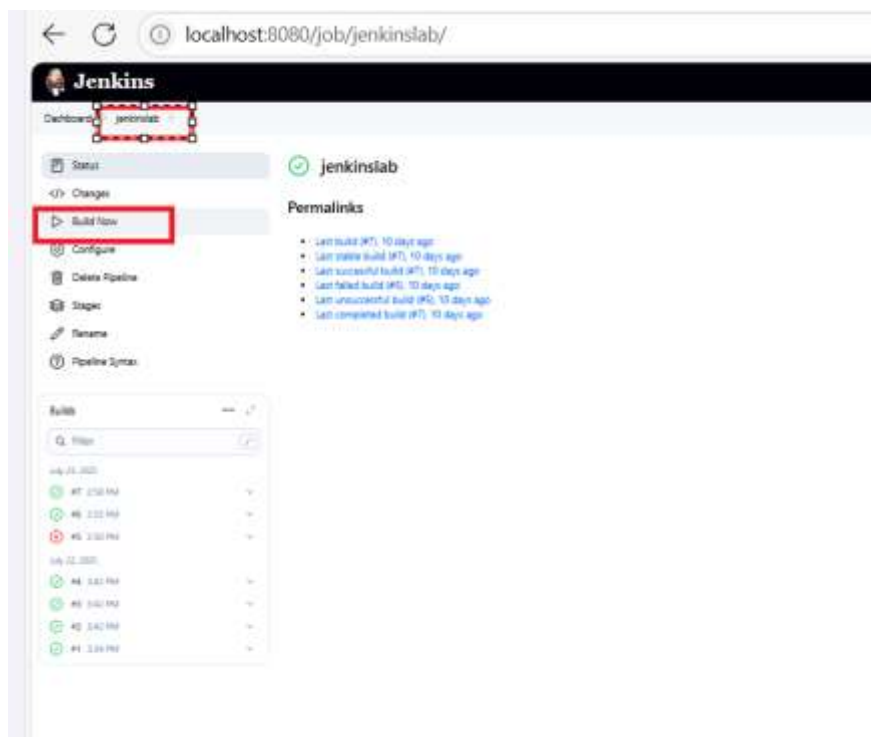
    post {
        success {
            echo "Build Successful!"
        }

        failure {
            echo "Build Failed!"
        }
    }
}

```



Go back to the created lab name and trigger the build by clicking on Build Now



Dashboard > jenkinslab > #7

Status

Changes

Console Output

Edit Build Information

Delete build '#7'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

✓ #7 (Jul 23, 2025, 2:58:26 PM)



Started by user [harshinikondapaka](#)



This run spent:

- 11 ms waiting;
- 5.4 sec build duration;
- 5.4 sec total from scheduled to completion.



Revision: 90e31e7a282d0238caf682b318e271512cf67

Repository: <https://github.com/harshiniadvik/jenkinslab>

- refs/remotes/origin/main



No changes.

The screenshot shows the Jenkins web interface. At the top, the Jenkins logo and name are displayed. Below the header, the breadcrumb navigation shows 'Dashboard > jenkinslab >'. On the left, a sidebar menu contains various options: Status (selected), Changes, Build Now, Configure, Delete Pipeline, Stages, Rename, and Pipeline Syntax. The main content area shows the 'jenkinslab' pipeline status as 'Success' (green checkmark). Below this, a 'Permalinks' section lists several links for different build types and statuses, all indicating they occurred '10 days ago'. At the bottom, a 'Builds' section displays a list of builds. Build #7 is highlighted with a red circle and a red line pointing to it. The build list shows builds for July 23, 2025, and July 22, 2025.

Jenkins

Dashboard > jenkinslab >

Status

jenkinslab

Permalinks

- [Last build \(#7\), 10 days ago](#)
- [Last stable build \(#7\), 10 days ago](#)
- [Last successful build \(#7\), 10 days ago](#)
- [Last failed build \(#5\), 10 days ago](#)
- [Last unsuccessful build \(#5\), 10 days ago](#)
- [Last completed build \(#7\), 10 days ago](#)

Builds

Filter

July 23, 2025

- ✓ #7 2:58 PM
- ✓ #6 2:55 PM
- ✗ #5 2:50 PM

July 22, 2025

- ✓ #4 3:42 PM

Check the console output

Dashboard > jenkinslab > #7

Status

</> Changes

Console Output

Edit Build Information

Delete build '#7'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

← Previous Build

✓ #7 (Jul 23, 2025, 2:58:26 PM)

Started by user [harshinikondapaka](#)

This run spent:

- 11 ms waiting;
- 5.4 sec build duration;
- 5.4 sec total from scheduled to completion.

git **Revision:** 90e31e7a282d0238caf682b318e271512cf67
Repository: <https://github.com/harshiniadvik/jenkinslab>

- refs/remotes/origin/main

</> No changes.

In the console output we will actually see the output of the java file which is in the next image

localhost:8080/job/jenkinslab/7/console

Jenkins

Dashboard > jenkinslab > #7

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build #7
- Timings
- Get Build Data
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces
- Previous Build

Console Output

```
Started by user harsh@jenkinslab
[Pipeline] Start of Pipeline
[Pipeline] end
Running on Jenkins in C:/Users/Admin/.jenkins/workspace/jenkinslab
[Pipeline] {
[Pipeline] withBsc
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout code)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
> git.exe rev-parse --revsion-git-dir C:/Users/Admin/.jenkins/workspace/jenkinslab/.git # timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/harshiniadvh/jenkinslab.git # timeout=10
Fetching upstream changes from https://github.com/harshiniadvh/jenkinslab.git
> git.exe --version # timeout=10
> git --version # 'git version 2.49.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/harshiniadvh/jenkinslab.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/main"[commit] # timeout=10
Checking out Revision 90e31e7a202d0238caf682b318e271512cf67307 (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 90e31e7a202d0238caf682b318e271512cf67307 # timeout=10
> git.exe branch -a -v --no-abbrev # timeout=10
> git.exe branch -D main # timeout=10
> git.exe checkout -b main 90e31e7a202d0238caf682b318e271512cf67307 # timeout=10
Commit message: "adding samplename.java"
> git.exe rev-list --no-walk 90e31e7a202d0238caf682b318e271512cf67307 # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] mvn
C:/Users/Admin/.jenkins/workspace/jenkinslab>mvn SampleName.java
[Pipeline] mvn
C:/Users/Admin/.jenkins/workspace/jenkinslab>java SampleName
Hello, my name is advhadiya
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] end
```

localhost:8080/job/jenkinslab/7/console

Dashboard > jenkinslab > #7

Previous Build

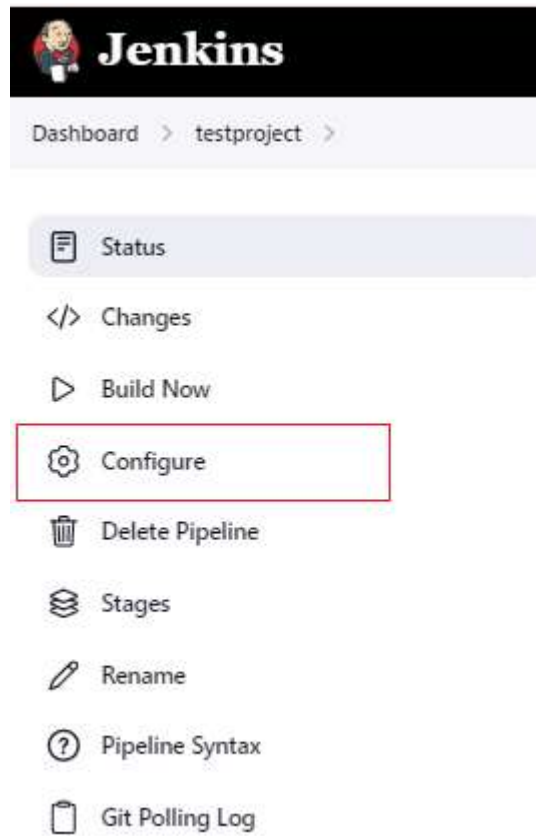
```
> git --version # 'git version 2.49.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/harshiniadvh/jenkinslab.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/main"[commit] # timeout=10
Checking out Revision 90e31e7a202d0238caf682b318e271512cf67307 (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 90e31e7a202d0238caf682b318e271512cf67307 # timeout=10
> git.exe branch -a -v --no-abbrev # timeout=10
> git.exe branch -D main # timeout=10
> git.exe checkout -b main 90e31e7a202d0238caf682b318e271512cf67307 # timeout=10
Commit message: "adding samplename.java"
> git.exe rev-list --no-walk 90e31e7a202d0238caf682b318e271512cf67307 # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] mvn
C:/Users/Admin/.jenkins/workspace/jenkinslab>mvn SampleName.java
[Pipeline] mvn
C:/Users/Admin/.jenkins/workspace/jenkinslab>java SampleName
Hello, my name is advhadiya
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] end
```

Result 1- We should be able to see the output of java in Jenkins console

Till here we are done with manual triggering of the build using Jenkins and now next we will try to do an automation . As soon as any new change is introduced to the java file Jenkins will automatically trigger the build

Steps to Set Poll SCM in Jenkins:

1. Open your Jenkins job.



2. Click on Configure.
3. Scroll down to the Build Triggers section.
4. Check the box labeled Poll SCM.
5. In the Schedule field, enter H/5 * * * *



6. Click apply & Save.

H/5 means Jenkins will poll every 5 minutes, but the H ensures that the polling happens at different minutes for different jobs, reducing the load on the server.

The other * symbols mean "every hour", "every day", "every month", and "every day of the week", respectively.

Result 2:

Modify the java program, add to the staging area, commit and push to github. Wait for 5 min and check Jenkins the job will be executed automatically